FYDP I POSTER / UNITED INTERNATIONAL UNIVERSITY · DEPT. OF CSE

Solving the Lorenz ODE System using Optimal ANN Architectures

Can a **plain feedforward ANN**, with no physics in the loss, learn a coupled nonlinear ODE system, and **which architecture does it best?**

SUPERVISOR
Dr. Muhammad Nomani Kabir
Professor, Dept. of CSE, UIU

COURSE TEACHER
Dr. Md. Motaharul Islam
Professor, Dept. of CSE, UIU

TEAM_PARADOX
Ikramul Hasan Moral · Md. Abu Bakar · Samiur Rahman Omlan · Fariha Islam · Md. Touhidul Islam

01 The Problem

Classical solver
Step-by-step, mesh-bound, re-runs per query

vs

Trained ANN
One direct map t → (x, y, z), mesh-free

Central question: can a plain ANN, trained only on data with no physics in the loss, learn this coupled system — and **which architecture does it best?**

02 Objectives

Trusted baseline
RK4 + SciPy, cross-checked

Search the space
Depth × width × activation

Accuracy vs cost
Error and run-time, together

Benchmark
Best ANN vs PINN literature

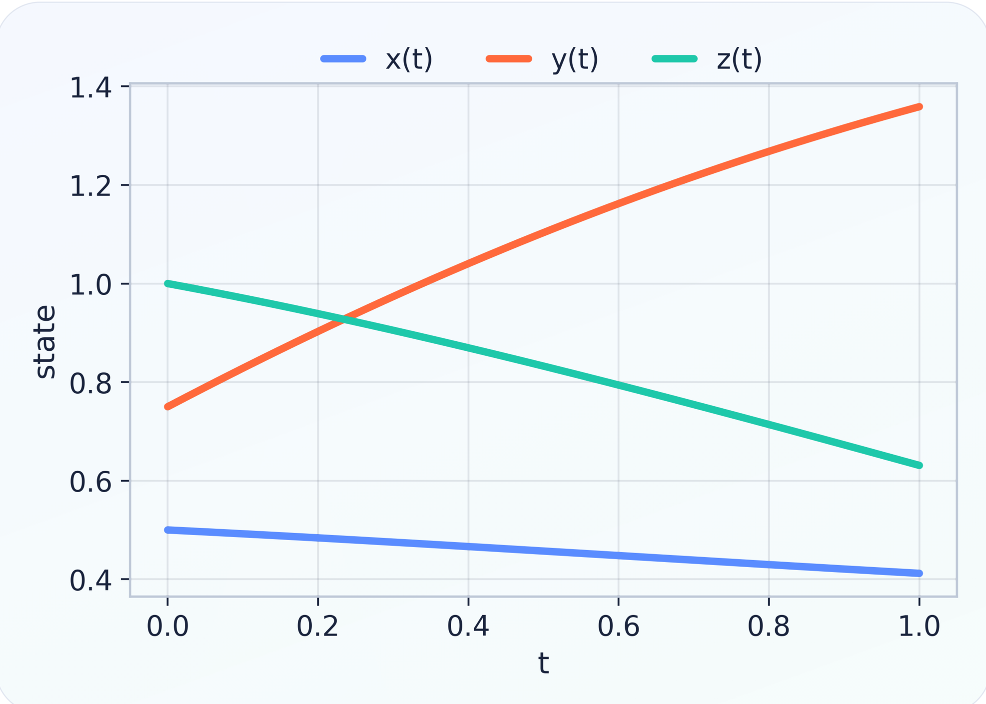
03 The System

REDUCED FORM · K = 2, L = 1

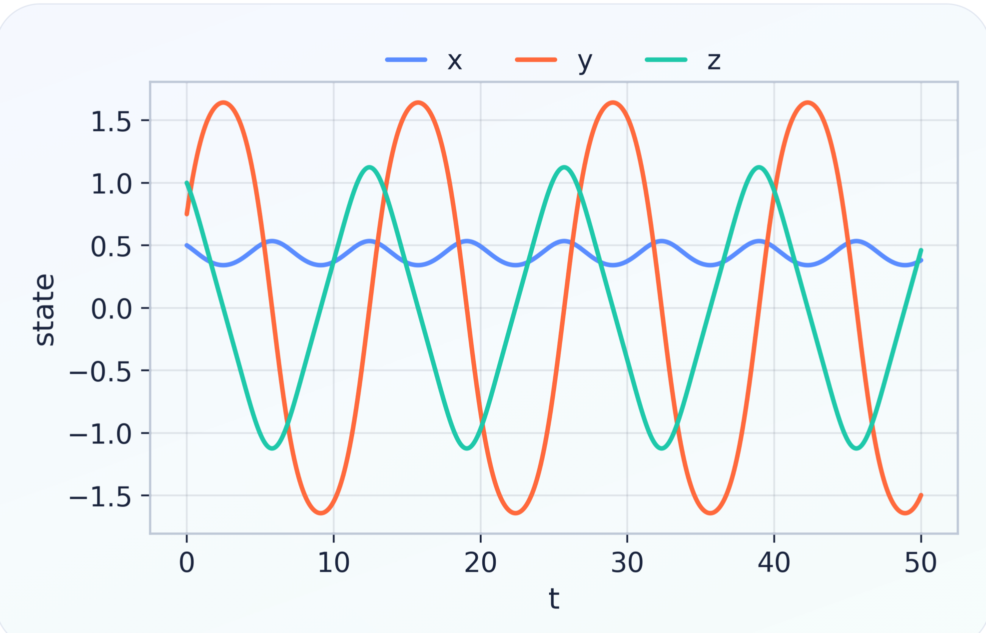
$$\dot{x} = -0.10\, y z$$
$$\dot{y} = +1.60\, x z$$
$$\dot{z} = -0.75\, x y$$

INITIAL	VALUES	INTERVAL
x_0, y_0, z_0	0.5, 0.75, 1	t ∈ [0,1]

Three coupled nonlinear ODEs: simple to state, hard to integrate accurately.

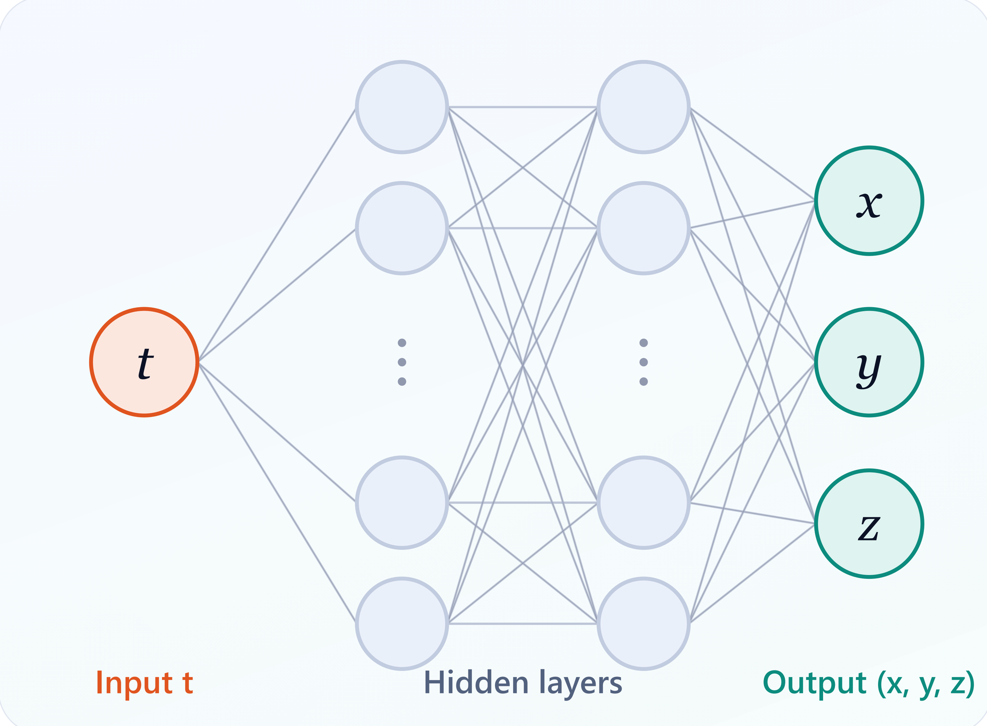


RK4 reference solution on t ∈ [0, 1]: the data the ANN learns.



Long-term behaviour on t ∈ [0, 50]: bounded, quasi-periodic orbits.

04 Network & Search Space



Depth (LAYERS): 1 · 2 · 3 · 4

×

Width (NEURONS): 20 · 50 · 100

×

Activation (NONLINEARITY): tanh · ReLU · sigmoid · GELU · Swish

= **60** candidate architectures

05 Two-Phase Study

PHASE 1
60
architecture search · Adam fixed

PHASE 2
9
optimizer sweep · top-3 nets

Architecture first, optimizer second; the two effects stay separable.

Pipeline

Generate
RK4 + SciPy reference

↓

Split
80 / 20 train-validation

↓

Train
69 controlled runs

↓

Select
best architecture

06 How We Score

MSE
mean sq. error

MAE
mean abs. error

Max error
worst case

Train time
fitting cost

Inference time
cost of a single query once trained

07 The Gap

PRIOR WORK	LORENZ-1960	PLAIN ANN	ARCH. SWEEP
Matthews & Bihlo	✓	✗	✗
Lagaris et al.	✗	✓	✗
Lau & Werth	✗	✓	✗
Mall & Chakraverty	✗	✓	✗
Ours · Team_ParaDox	✓	✓	✓

No prior study sweeps plain-ANN architecture on Lorenz-1960.

08 Baseline Verified

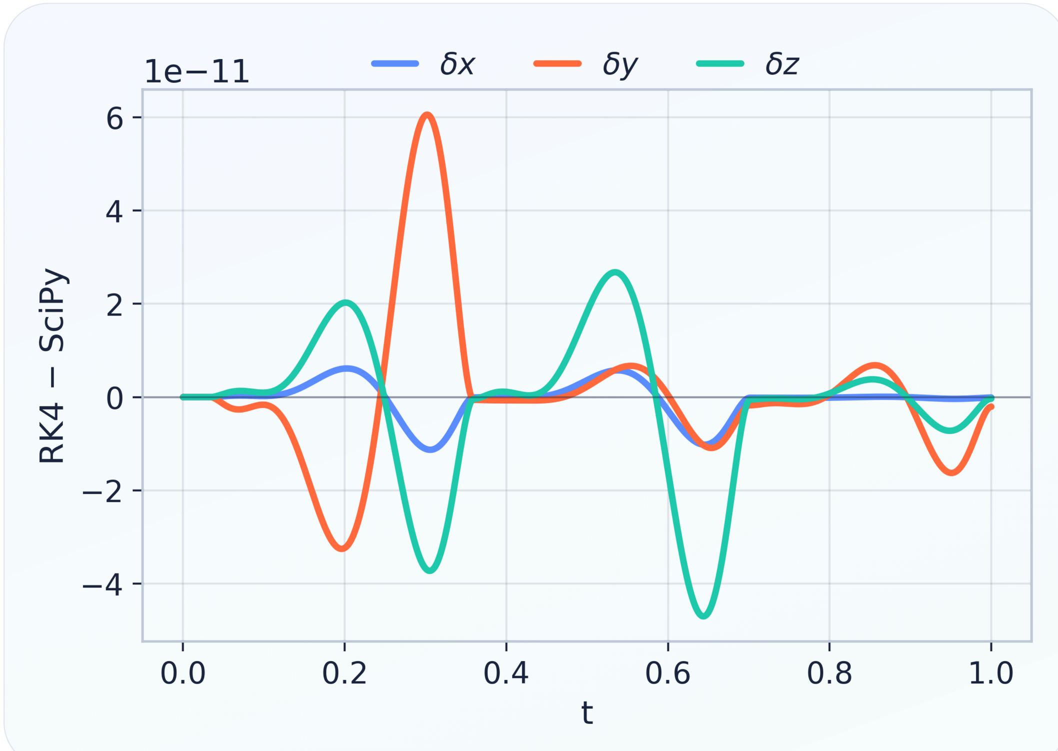
target < 10^{-10}

1.8×10^{-22}
solver-agreement MSE (RK4 vs SciPy)

6.06×10^{-11}
max pointwise disagreement

✓ target beaten by ten orders of magnitude

Two independent solvers agree before any ANN is trained.



Pointwise RK4 - SciPy disagreement across the trajectory.

09 Status & Next

FYDP-I · done
17-paper review, verified **dual-solver baseline**, locked **69-run** design

FYDP-II · next
Run the experiments, repeat seeds, **benchmark against PINNs**

No claims yet. Accuracy results follow once FYDP-II runs.